

Page 3: Project Design Phase

1. Introduction

The **Project Design Phase** outlines the structural, architectural, and visual framework of **Smart Budget AI**. This phase transforms the functional specifications from Phase 2 into technical blueprints, system diagrams, database schemas, and user experience (UX) workflows.

2. High-Level System Architecture

The application follows a **Microservices-Based Cloud Architecture** to ensure modularity, scalability, and security.

- **Presentation Layer (Frontend):** Cross-platform mobile/web application built with React Native / Flutter providing real-time data visualization and interactive dashboards.
- **API Gateway Layer:** Manages authentication, rate limiting, request routing, and load balancing between the frontend and backend services.
- **Application Services Layer (Backend):**
 - **User & Auth Service:** Manages registration, session tokens, MFA, and user profile preferences.
 - **Transaction Engine:** Handles manual entries, statement parsing, and financial ledger management.
 - **OCR & Ingestion Service:** Extracts raw text from uploaded receipt images and structures payment details.
- **AI & Machine Learning Engine:**
 - **NLP Categorizer:** Classifies transactions into predefined or custom categories.
 - **Time-Series Forecaster:** Predicts future cash flows using ML models (e.g., ARIMA, Prophet).
 - **Conversational AI Module:** Connects to LLMs to answer natural language financial questions.
- **Data Storage Layer:** Relational DB for core financial records, NoSQL DB for unstructured logs/OCR cache, and Redis for high-speed session management.

3. Database Schema Design (Entity-Relationship Highlights)

- **Users:** user_id (PK), full_name, email, password_hash, currency, created_at
- **Accounts / Balances:** account_id (PK), user_id (FK), account_type (Bank, Cash, Wallet), current_balance
- **Categories:** category_id (PK), category_name, icon_url, is_system_default
- **Transactions:** transaction_id (PK), user_id (FK), account_id (FK), category_id (FK), amount, transaction_type (Income/Expense), merchant_name, timestamp, is_flagged_by_ai
- **Budgets & Goals:** goal_id (PK), user_id (FK), target_amount, current_saved, deadline, status

4. User Interface (UI) & User Experience (UX) Flow

- 1. **Onboarding & Authentication:** Splash screen → Login / Register → Currency & Salary Setup.
- 2. **Dashboard / Home:** Real-time monthly balance overview, visual spending breakdown (charts), recent transactions, and AI budget warning banners.
- 3. **Smart Ingestion Flow:** User snaps photo of receipt → OCR processing screen → AI suggested category preview → One-tap user confirmation.
- 4. **AI Financial Assistant (Chatbot Screen):** Conversational chat UI with prompt suggestions (e.g., *"Show my coffee spending this week"*).
- 5. **Analytics & Budget Planning:** Interactive trends analysis, savings goal progress, and predictive projections for upcoming months.

5. Key System Modules & Component Interaction

